

Stage 2: Results and Analysis

Single-hop Causal Head Map in OLMo-2-7B

Daniella Simonovsky and Max Golubkov

September 2026

1 Run and validation

Run `stage2_v1` completed successfully as Slurm job 888691 in 1h14m42s with exit code 0. Three persistent replicas used six GPUs in total, with two GPUs per replica. The run used the base `allenai/OLMo-2-1124-7B` model, revision `7df9a82518afdecae4e8c026b27adccc8c1f0032`, and the same data, behavioral baseline, discovery split, and Stage-1 artifacts as the observational scan.

All mandatory gates passed independently on all three replicas. The behavioral runner drift, self-patch drift, self-attribution effect, and corrupted-input embedding endpoint drift were zero on the checked examples. The gate also covered shared-answer-prefix cases and exact versus gradient previews. These checks establish consistency of this implementation on the tested inputs; they do not establish scientific usefulness of any head.

The output is complete: all 178 clean/corrupted discovery pairs have finite screen and extension files; 40 pairs have exact effects for all 1,024 heads; and all 178 pairs have exact effects for the 92-head extension set. The raw arrays reproduce `head_effects.csv`. Model, data, baseline, Stage-1, and code hashes match the saved manifest.

2 Intervention and outcome measure

For each family and fact order, let x_c be the clean prompt and x_r its token-aligned corrupted twin. The corruption swaps the two answer-side names in the facts, so that the clean distractor becomes the corrupted answer. If the two answer strings share a token prefix c , the metric is evaluated at their first divergent token:

$$M(x) = \text{logit}(y_c \mid x, c) - \text{logit}(y_r \mid x, c). \quad (1)$$

The sign is fixed to the clean answer on both inputs. Thus, $M(x_c) > 0$ and $M(x_r) < 0$ indicate that the model tracks the changed fact. Four of the 178 pairs had a non-empty shared answer prefix.

For head h , exact patching replaces its slice of the input to the attention output projection with the activation from the corrupted run at every original prompt position. Shared answer-prefix positions are recomputed. Define

$$\Delta_h = M(x_c; a_h \leftarrow a_h^r) - M(x_c), \quad I_h = -\Delta_h. \quad (2)$$

We report I_h as *causal importance*: a positive value means that the patch moves the metric toward the corrupted answer. It is measured in logits. It is not an accuracy difference, a sequence-likelihood difference, or an additive decomposition of the clean–corrupted gap.

Across the 89 discovery families, averaging the two orders within each family, the mean clean metric was 5.680 (95% family-bootstrap interval [5.338, 6.030]), while the corrupted metric with the clean

sign fixed was -5.491 ($[-5.850, -5.141]$). The mean gap was 11.170 ($[10.632, 11.725]$). Every clean pair had positive M and every corrupted pair had negative M . The intervention contrast is therefore behaviorally strong and sign-consistent.

Intervals in this report are descriptive percentile bootstrap intervals from 20,000 resamples of whole families (seed 20260915), after averaging the two orders. They were computed after observing the Stage-2 results and are not additional pre-registered hypothesis tests.

3 The first-order screen is well calibrated

The first-order estimate contracts the clean-to-corrupted head-output change at all original prompt positions with the gradient of M at the clean run. On the 40-pair exact subset, using the same pairs for both methods, its signed head-ranking Spearman correlation with exact patching was 0.945. The correlation of absolute mean effects was 0.921, and the pair-by-head correlation was 0.903. This passed the pre-registered threshold of 0.7, so the five-step input-embedding IG fallback was correctly not used.

Two further comparisons separate approximation error from family-sampling variation. Comparing the attribution mean over all 178 pairs with the exact mean over only the 40-pair subset gives a lower correlation of 0.698. In contrast, among the 92 extension heads, attribution and exact patching evaluated on the same 178 pairs correlate at 0.990. The lower cross-sample value is therefore not evidence that the screen failed; it mixes approximation error with a change in the evaluated families.

4 Causal map

Table 1 reports the strongest heads in the all-family exact extension. The extension set was selected on the discovery data using RI, attribution, fixed Stage-1 examples, and random controls. The intervals therefore describe effect stability within this run; they are not selection-adjusted confidence statements.

Head	RI	Attribution	Exact I_h	Gap fraction
L17H1	.0094	3.362	5.701 [5.294, 6.112]	.510
L18H19	.0182	2.785	3.072 [2.867, 3.276]	.275
L27H6	.0000	2.640	3.071 [2.749, 3.390]	.275
L18H18	.0151	2.226	2.704 [2.503, 2.907]	.242
L21H18	.0096	1.480	1.704 [1.588, 1.822]	.153
L25H17	.0116	1.183	1.267 [1.190, 1.342]	.113
L22H5	.0155	.950	.983 [.904, 1.063]	.088
L16H21	.0240	.695	.925 [.808, 1.049]	.083
L16H1	.0137	.812	.868 [.766, .972]	.078
L21H6	.0067	.662	.702 [.636, .770]	.063

Table 1: Strongest exact effects among the 92 extension heads. Gap fraction is I_h divided by the mean 11.170-logit clean–corrupted gap. It must not be summed across heads because single-head patching effects can interact.

L17H1 is unusually strong: replacing this one head moves the metric by 5.701 logits on average, approximately 51% of the full clean–corrupted gap. Its effect had the expected sign in every family. L18H19, L27H6, and L18H18 each account individually for approximately one quarter of the gap, again with the expected sign in every family. These magnitudes establish intervention sensitivity, not four independent shares of a decomposition.

The all-head exact map used a randomly chosen set of 20 whole families, both orders. Its top four heads were the same four shown above, and their effect magnitudes closely reproduced the 89-family extension: L17H1, L27H6, L18H19, and L18H18 had subset effects 5.742, 3.222, 3.080, and 2.768. The large effects are therefore not artifacts of reporting only the all-family extension.

Across all 1,024 heads, the order-0 versus order-1 exact ranking correlation was 0.461. This value is dominated by the many heads whose effects are near zero. Within the selected 92 heads the correlation was 0.734, and among the overall top 25 it was 0.656; 17 heads appeared in both order-specific top-25 lists. All ten heads in Table 1 had positive mean importance in both orders. Thus the main effects are not confined to one layout, while weaker-head ordering is less stable.

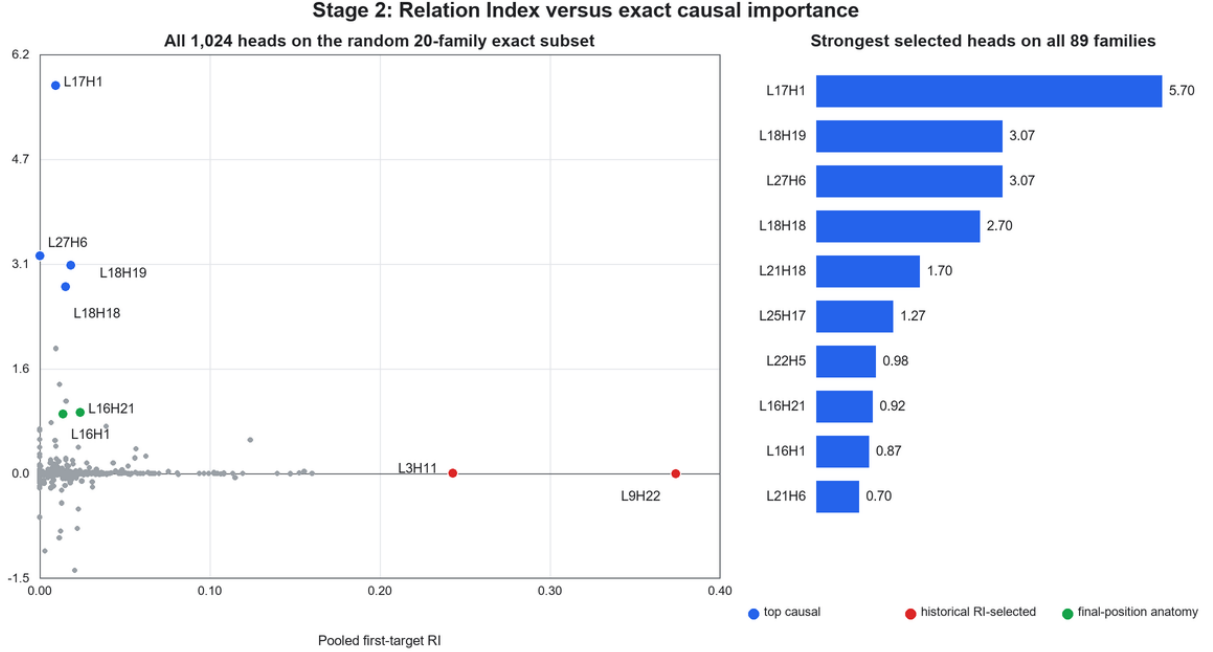


Figure 1: Stage-2 causal map. Left: pooled first-target RI versus exact causal importance for all 1,024 heads on the random 20-family subset. Right: strongest heads among the 92-head exact extension on all 89 discovery families. The historical RI-selected heads lie near zero causal importance, while the largest causal effects have low RI.

5 RI has weak correspondence with causal importance

The Spearman correlation between pooled first-target RI and exact importance over all heads was 0.135 on the random exact subset. A family bootstrap gave a descriptive stability interval of $[0.063, 0.168]$. The last-target sensitivity score performed similarly ($\rho = 0.132$). These small positive associations do not support RI as a useful causal ranking for this task.

Only one of the exact top-25 heads appeared in the supported RI-first top 25, and only one appeared in the RI-last top 25. By comparison, 17 of the exact top-25 appeared in the attribution top 25. The overlap result agrees with the rank correlations: the gradient screen tracks exact effects, whereas the RI ranking mostly selects different heads.

The historical pooled rule had selected L3H11 and L9H22. On all 89 families, their exact importance was -0.0022 ($[-0.0067, 0.0024]$) and -0.0020 ($[-0.0065, 0.0027]$), with the expected sign in only 47% and 42% of families. Their median importance was -0.0021 , compared with 0.0002 for the 25 random extension controls. These heads therefore have no detectable effect under this single-head, all-prompt-position patching experiment.

This result sharpens the Stage-1 event analysis. L3H11’s RI events were all previous-token attention from the same period/newline token, and L9H22’s events were all self-attention. More than 99% of their summed first-target scores came from demonstrations. Stage 2 now shows that the large observational scores do not translate into movement of the answer metric when their contextual head activations are transplanted.

6 Pre-registered audit verdicts

The earlier methodology defined three non-exclusive verdicts. The available evidence supports the following reading.

Incomplete: supported. The lenient 99th percentile of the historical random-token control means was 0.1374. Of the 102 heads in the top decile of positive exact effect, 101 fell below this threshold and 24 had RI exactly zero. Most decisively, L27H6 ranked second in exact importance but had RI zero. Thus a large causal effect can lie below even the lenient RI threshold; this conclusion does not depend on choosing a stricter cutoff after observing Stage 2.

Misleading: descriptively supported. The pre-registered criterion asked whether the median causal effect of RI-passing heads exceeded that of random heads. It did not: the two historical RI-passing heads had median importance -0.0021 , versus 0.0002 for the random controls. The historical RI threshold itself was a descriptive cross-head percentile rather than a calibrated per-head significance test, so this verdict should be reported with that qualification.

Sound: not supported. The methodology described soundness as systematically larger effects for RI-passing heads than for layer-matched non-passing heads, but did not fully operationalize the comparison. Only two heads passed the historical rule, one of which ranked high within its early layer on the 20-family subset but had a tiny effect that vanished on the full 89-family extension. The present data therefore do not support soundness, but we do not turn the underspecified criterion into a new post-hoc significance test.

These verdicts concern the implemented RI adaptation and causal experiment on this dataset. They are not a universal rejection of semantic induction heads or of attention-based diagnostics.

7 Two distinct ways RI misses important heads

The combination of Stage-1 events and Stage-2 effects suggests at least two failure modes.

QK-selection miss. L17H1 had only two scored Stage-1 QK passes across the full scan, yet it was the strongest causal head by a large margin. Its mechanism therefore rarely satisfies the specific condition that the annotated source token be the unique dominant attention target by a factor greater than 2.2. It may distribute attention, retrieve through a different position, or operate on information already moved by upstream components.

Raw-embedding OV miss. L27H6 had 85 scored QK passes but pooled RI zero, despite an exact effect of 3.071 logits. This separates attention eligibility from the embedding-level OV score. A head can be causally important even when $e_j W_V^h W_O^h W_U$ does not promote the annotated target token. Its actual contextual input may carry information absent from the raw token embedding, or its downstream effect may be indirect rather than a direct target-token write.

L16H1 and L16H21 provide a third informative pattern. Stage 1 found 223 and 230 dominant-attention events, respectively, from the final prompt token to test-fact sources across 76 and 79 families. Stage 2 found exact effects of 0.868 and 0.925 logits. This agreement makes them promising query-position retrieval candidates. However, their low pooled RI shows that the full RI definition still under-ranks them, plausibly because its raw-embedding OV term does not measure their contextual output.

L16H4, which had the smallest raw matched-target calibration p value in Stage 1 and 63 final-position source-attention events, had a positive but much smaller exact effect: 0.0257 logits ([0.0190, 0.0328]), approximately 0.2% of the clean–corrupted gap. Dominant source attention and a weak target preference can therefore coexist with only a small causal effect on the answer contrast.

8 Interpretation and next mechanistic questions

Stage 2 establishes a reliable causal screen under a precisely specified intervention. It provides strong evidence that the implemented RI definition is not a complete causal account of the model’s single-hop relational behavior. It also identifies a compact set of robust intervention-sensitive heads, concentrated mainly in layers 16–27, for the next stage.

The result does *not* yet identify a circuit. Because each patch replaced a head at all original prompt positions, it does not localize the effect to the question, relevant fact, or answer-prediction position. A large effect may reflect semantic relation processing, entity binding, name copying, or another feature that differs between the token-aligned twins. Single-head effects also do not establish necessity under redundancy, edges between heads, or additive composition.

Stage 3 should therefore begin with L17H1, L18H19, L27H6, L18H18, L16H1, and L16H21, while retaining the original RI heads and random controls. The first questions are:

1. At which prompt positions does each exact effect arise?
2. Where does each head attend on clean and corrupted twins, and does the attention track the query-relevant fact rather than a fixed location?
3. What does the head’s *actual contextual output*, rather than its raw embedding OV projection, write into the residual stream?
4. Which upstream components supply its queries, keys, and values, and which downstream components consume its output?
5. Do pairwise and group interventions reveal redundancy or nonlinear interactions that single-head patches cannot detect?

Only after these position- and edge-level tests should the components be called a semantic circuit or evaluated for minimal-circuit faithfulness on the held-out validation families.

9 Conclusion

The causal map changes the Stage-1 picture. The two heads selected by pooled RI have essentially zero exact effect, while several low-RI heads have large, stable, and correctly signed effects. The first-order screen is strongly validated against exact patching, so this contrast is not explained by a failed approximation. Within the implemented task and patching scope, RI is incomplete and descriptively misleading as a causal selection rule. The main positive result is not merely that RI fails: Stage 2 identifies a reproducible set of mid-to-late-layer heads whose mechanisms can now be tested at position and edge resolution.